

BioSpecInfo

Dossier completo

Samuele Pio Provenzano

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Raccolta integrale dei documenti del progetto, nell'ordine di numerazione. Ogni documento è disponibile anche come PDF separato.

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BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	RDKit cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as PWA	Spectra Centre : IR, ¹ H/ ¹³ C-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and 3D (three.js, 3Dmol.js) rendering	Organic Chemistry : 25 interactive modules
Service Worker offline-first, versioned	Retrosynthesis : 46 strategies + animated 2D/3D mechanisms

Technical architecture	Content & capabilities
Zero copleft dependencies (MIT/BSD-3 only)	3D Drug Atlas: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **¹H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ± 0.5 ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm⁻¹** (90% within ± 30 cm⁻¹)
- Interactive performance **30-60 fps** from iPhone/iPad to Raspberry Pi 4

3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, Brenk/PAINS structural alerts, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

4. Technology stack

HTML5 · ES6/ESNext · WebAssembly (RDKit MinimalLib, sql.js/SQLite) · WebGL (three.js, 3Dmol.js) · SVG / Canvas 2D · Service Worker / PWA · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.
1. The reported validation data (MAE 0.31 ppm for ^1H -NMR vs SDBS; MAE 11 cm^{-1} for the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.
1. The architecture is **local-first**: no user personal data is transmitted to author-controlled servers (alignment with GDPR principles).

This is an **author's declaration**, not a certification issued by a third party.

6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
 2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
 3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
 4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.
- _Software license: proprietary — "All rights reserved" (see [LICENSE](#))._

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	<code>samupropiol-ship-it/BioSpecInfo-v11</code>

1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

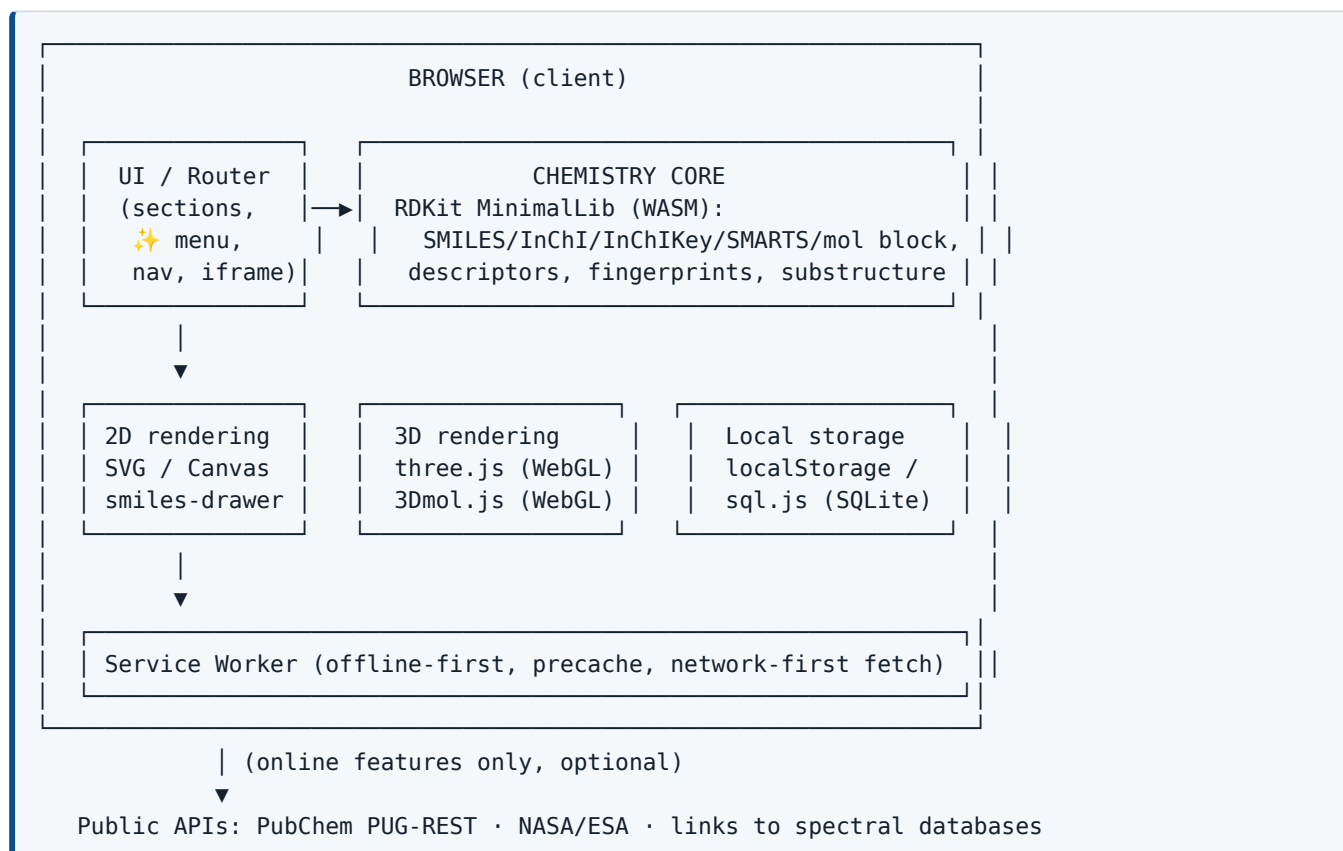
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

2. File structure

File	Role
<code>index.html</code>	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
<code>chimorga.html</code>	Organic Chemistry course (25 modules).
<code>astro.html</code>	3D Astrochemistry (three.js).

File	Role
pro.html, rdkit_lab.html, accademia.html, simulazioni.html, guidaret.html, file_manager.html, Biochimica_Guida_Definitiva.html, sr_completo.html, sr_essenziale.html, download.html, changelog_tesi.html	Thematic sub-apps loaded in iframes with <code>?v=timestamp</code> cache-busting.
manifest.json	PWA configuration (icons, theme, standalone).
sw.js	Service Worker: <i>network-first</i> strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
RDKit_minimal.js / RDKit_minimal.wasm	Cheminformatics engine (RDKit MinimalLib).
3Dmol-min.js	3D viewer for molecules/macromolecules.
three.min.js, three_bloom.js, gltf_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
smiles-drawer.min.js	2D depiction of SMILES.
lib/sql-wasm.js / lib/sql-wasm.wasm	SQLite in WebAssembly (SQL playground).
models/, textures/	3D assets (organs, écorché, planets) and textures.

3. Component architecture



3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframes`.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS, MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP (Wildman-Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp³, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg` , `get_svg_with_highlights`) and 2D coordinate generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D (molecule editor, CIP stereochemistry viewer), smiles-drawer.
- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito / blocked-storage scenarios).
- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.

- **Predicted spectra** as sub-panels: IR, ^1H -NMR, ^{13}C -NMR, MS with ****isotope pattern (IUPAC abundances) and adduct masses**** ($[\text{M}+\text{H}]^+$, $[\text{M}+\text{Na}]^+$...), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET**: Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry ([chimorga.html](#))

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground ([sql.js](#)), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + [SharedArrayBuffer / Atomics](#) .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR; hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
- Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.

6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real

3D models.

- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	<code>node --check</code> on the critical JavaScript blocks on every change.	✓ In use
End-to-end (E2E) testing	Automation with Playwright on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: 0 non-network errors).	✓ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✓ In use
Manual cross-device testing	Testing on real devices (see §4).	✓ In use
Automated unit-test suite (Vitest/Jest)	—	 Roadmap
Coverage ≥ 90%	—	 Roadmap

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	SDBS (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	0.31 ppm
Signals within ± 0.5 ppm	89%

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	NIST Chemistry WebBook
Mean Absolute Error (MAE)	11 cm⁻¹
Bands within ± 30 cm ⁻¹	90%

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol* $\rightarrow$ *C₂₇H₄₆O*, 4 rings, 8 stereocentres.

Benchmark data source: measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30-60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30-60 fps

Size of the main `index.html` file: $\approx$ **4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

- Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.
 - Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure-spectrum relationships, not to replace certified experimental data.
 - Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.
 - Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.
-

6. Release acceptance criteria

- ✓ `node --check` passes without errors on the modified modules.
 - ✓ Playwright E2E: 0 non-network JS errors.
 - ✓ No visible regression in the touched sections.
 - ✓ Service Worker cache version bumped.
 - ✓ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
 - ✓ `verifica_guida`: whatever the documentation promises exists in the code.
-

7. Verification battery — state at version `bsi-v172`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	<code>verifica-farmaci</code> , <code>test_spettri</code> , <code>test_assi</code> , <code>test_assi_canvas</code> , <code>test_costanti</code> , <code>audit_dati</code>	structure vs molecular weight, functional-group recognition, axis conventions, physical constants

Family	Benches	Subject
AI agent	test_ko, test_404, test_503, test_firma, test_attesa, test_attesalunga, test_tetto, test_nucleo, browser_ko, browser_prova	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	audit_stabilita, audit_promesse, audit_quota, test_filemanager, test_doppioinvio, test_sw, test_visore3d	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	browser_reset, browser_proxy, browser_proxyui, browser_rdkit, browser_lab, browser_frontiera, test_aggiorna, test_guidaproxy	panels, proxy, RDKit laboratory, updates
Consistency	verifica_guida	documentation ↔ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	87 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, +0 on the four that follow
Survival of exhausted storage	localStorage.setItem forced to throw	10 pages out of 10 stay operational
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in 3,507 ms (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to 1

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.
- Probing `localStorage` exhaustion with a one-byte write proves nothing: after saturation, one byte always finds room.
- The ticks of an axis drawn on `<canvas>` are not readable as text: they must be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.

Security, Privacy & Compliance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Model	Client-side / Local-First PWA

1. Privacy & data protection (GDPR)

1.1 Local-First by design

- **No account, no login, no advertising tracking.**
- All user data (drawn molecules, preferences, quiz history, study progress) is stored **exclusively locally** on the device (`localStorage` / IndexedDB / `sql.js`), with no transmission to author-controlled servers.
- **There is no application backend** collecting or processing personal data.

1.2 Data handling and GDPR alignment

- **Data minimisation:** the app requires no personal data to function.
- **No transfer to the author's third parties:** data never leaves the device for infrastructure controlled by the author.
- **Optional, transparent external calls:** only upon a user action (searching a molecule, opening a 3D model) does the app query public third-party APIs (PubChem, NASA/ESA, spectral databases). Such requests contain **only the chemical identifier** (e.g. SMILES/name) and **no personal data**. Use of these services is subject to their respective privacy policies.
- **Local right to erasure:** the user can delete data by clearing the site storage in the browser.

Note. This alignment describes the local-first design properties that favour GDPR compliance. It does not constitute a legal certification; for regulated contexts a dedicated DPIA assessment is recommended.

2. Application security (OWASP)

2.1 Attack surface

Being a **static application with no backend**, entire classes of server-side vulnerabilities are absent by construction (SQL injection on a server DB, server-side RCE, server authentication/session, IDOR on proprietary APIs).

2.2 Input sanitisation

- Parsing of **SMILES / SMARTS** strings and molecular files is delegated to

RDKit MinimalLib (WASM), which validates the structures and rejects malformed input by returning invalid molecules rather than executing them.

- Strings coming from user input or external APIs, when inserted into the DOM, are treated as untrusted. Their textual value (names, notations) is shown as text content/ `textContent` or with escaping of the `&` `<` `>` characters.

2.3 Runtime robustness

- **localStorage guard**: every storage access is protected by `try/catch` with an in-memory fallback, so that incognito mode or blocked storage cannot halt execution (a historical regression that has been fixed).
- **Controlled degradation**: panels never silently stay empty; errors show a message and, where applicable, a "Retry" button.
- **Sub-app isolation** in `iframe`s.

2.4 Content Security Policy (deployment recommendation)

Since the app can be published on static hosting, it is recommended to serve it with security headers at the hosting/CDN level:

- Restrictive `Content-Security-Policy` (self script/style + WASM),
`frame-ancestors`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`.
- `Cross-Origin-Opener-Policy` / `Cross-Origin-Embedder-Policy` if `SharedArrayBuffer` and parallel Web Workers are enabled in the future (HPC roadmap).

3. Integrity and reproducibility (GAMP 5 alignment)

For use in pharmaceutical/regulated contexts, the architecture is **compatible** with GAMP 5 principles:

- **Determinism**: the cheminformatics computations (RDKit) are deterministic and reproducible for the same input.

- **Version traceability:** code versioning on Git, versioned Service Worker cache, changelog.

- **Method transparency:** the heuristic predictors are documented with their sources and limitations (see the *Validation Report*).

Again: this is **alignment with the principles**, not a formal qualification (IQ/OQ/PQ), which would require a dedicated validation process at the using organisation.

4. Summary

Aspect	Status
Personal data transmitted to author's servers	None
Backend / server database	Absent
Molecular input validation	RDKit WASM
DOM output escaping	Yes (untrusted content)
Storage guard / error degradation	Yes
CSP / security headers	Recommended at hosting level
Offline operation	Yes (Service Worker)

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	RDKit_minimal.js , RDKit_minimal.wasm	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	BSD-3-Clause
3Dmol.js	3Dmol-min.js	3D viewer for molecules/macromolecules (WebGL)	BSD-3-Clause
three.js	three.min.js , three_bloom.js , glTF_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	MIT
SmilesDrawer	smiles-drawer.min.js	2D depiction of SMILES strings	MIT
sql.js (SQLite compiled to WASM)	lib/sql-wasm.js , lib/sql-wasm.wasm	Client-side SQL playground	MIT

The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms

Service	Use
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

3. Intellectual property of the proprietary code

- BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine, the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.
- No copyleft-licensed code** is embedded in the proprietary source: the author retains full licensing freedom over their own code.

3.1 Project license

The repository includes a **proprietary — "All rights reserved"** [LICENSE](#) file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	None
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project LICENSE file	 Present — proprietary "All rights reserved"

AI Agent Architecture "Spectra" — BioSpecInfo

Field	Value
Project	BioSpecInfo — Spectra component (agentic AI copilot)
Author	Samuele Pio Provenzano
Thesis supervisor	Prof. Savino Longo — University of Bari Aldo Moro
Component	<code>bsi-ai-hub.js</code> — 6,254 lines, zero runtime dependencies
Type	Multi-provider conversational agent with client-side tool execution
Repository	<code>samupropio1-ship-it/BioSpecInfo-v11</code>
Documented version	Service Worker <code>bsi-v172</code>

1. Executive summary

Spectra is an **agent**, not a thin wrapper around a language model. The difference is measurable: the model does not answer quantitative questions from memory — it **invokes deterministic tools** that compute the result, and the control loop lets it **chain up to 10 calls**, verifying each step before answering.

The architecture addresses three problems that separate a usable agent from a demo:

1. **Grounding** — a fabricated chemical value can be dangerous. 35 tools cover computation, public databases and the application's own datasets; the system prompt explicitly forbids quoting numeric values from memory.
1. **Transparency** — the model's reasoning is streamed live and stored alongside the answer, together with the list of tools actually invoked. The agent's work is auditable after the fact.
1. **Provider portability** — a single tool registry is translated into the three function-calling formats in use today (OpenAI-compatible, Anthropic, Gemini), so the agent behaves identically across ten model configurations, four of them free.

2. Architecture

2.1 Overview



There is no backend. API keys stay in the browser's `localStorage` and travel only to the selected provider: no data passes through BioSpecInfo servers, which do not exist.

2.2 Agentic loop

Each round: send the conversation history (last 40 turns) plus the system prompt and the schema of all 35 tools → receive the streamed response → if it contains tool calls, execute **all** of them → rebuild the assistant turn in the provider's native format → repeat.

Three exit conditions: no tool calls (final answer), `MAX_ROUNDS = 10` exhausted, or user interruption.

2.3 Provider-neutral tool registry

Tools are declared once in JSON Schema and translated at runtime:

Family	Required shape
Anthropic	<code>{name, description, input_schema}</code>
Gemini	<code>[{functionDeclarations: [...]}]</code>
OpenAI-compatible	<code>{type:"function", function:{...}}</code>

The same applies to messages: turns containing tool calls are serialised into each provider's native form and kept in a `_native` field, so a conversation stays coherent even when switching models.

2.4 No hard-coded model names

A model name written into the code is a time bomb: it works until the provider retires that model, and then the application stops with a 404 the user cannot fix. It has happened twice in this project:

Provider	Error	When
Google	<code>models/gemini-1.5-flash</code> is not found for API version <code>vlbeta</code>	removed from <code>vlbeta</code>
Groq	The model <code>llama-3.3-70b-versatile</code> does not exist or you do not have access to it	deprecated 17/06/2026

On some services the problem is structural: free model ids change constantly, by design. (That is why OpenRouter was later dropped from the list, see 2.4.)

The fix is not to write the new name — that would be the same bomb with a longer fuse — but to **remove the name** and ask the API, which knows what exists *for that key*. This also resolves the ambiguity in Groq's message: "does not exist" **or** "you do not have access" are different cases, and a per-key listing tells them apart.

Every configuration with `modelIdCandidates` starts at `model: null` and is resolved at runtime. The two families use different strategies because their names are different:

Gemini — version scoring. Names follow a regular scheme (`gemini-<major>.<minor>-<family>`), so they can be ordered: newest version first, `flash` family preferred, with penalties for experimental, `preview` and `-lite` variants; models without `streamGenerateContent` are discarded, as are non-conversational ones (embedding, image, native audio, Live API). When Gemini 3.0 ships it will be picked automatically.

OpenAI family — candidates in preference order. Here names are not comparable to each other (`openai/gpt-oss-120b` versus `qwen/qwen3.6-27b`) and automatic scoring would choose badly. `GET /models` is queried and the **first candidate that still exists** wins: the order of the list *is* the preference, and the listing serves to skip the ones that vanished. Only if no candidate survives does a generic score run over the available models — discarding transcription, speech, embedding and guard models, and preferring the larger model all else being equal.

Three fallback levels in both cases: model listing → candidate probing → first candidate, letting the real error from the generation call speak instead of inventing one.

The choice is cached for seven days per provider, keyed to a 32-bit FNV-1a fingerprint of the API key (different keys see different catalogues; the key itself is never duplicated). If the model is retired *while* cached, the `404` — or a `400` whose text mentions the model, as some providers return — invalidates the cache, re-resolves and retries **once**: a flag prevents an infinite loop when the new model fails too.

Anthropic is the deliberate exception: its models are paid, explicitly chosen by the user, and deprecated with long notice.

The three free services, chosen one by one

Free quality is not all alike, and the difference matters: an 8-billion-parameter model does not hold up on a multi-step physical-chemistry problem. The list has been **shortened**, not extended — two services were removed because they made answers worse, not better.

Service	Models	Its role
Groq	GPT-OSS 120B, Qwen3	The workhorse: 131K context, ~30 requests/minute
Google Gemini	Gemini Flash	Up to 1M context: whole documents and PDFs
Z.AI GLM	GLM-4.7-Flash	Free with no expiry : the reserve still there when credits are gone

Removed. *GitHub Models*: it was here, and was removed because GitHub **retired it entirely on 30 July 2026** — playground, catalogue and inference API switched off for everyone. It had been added two days earlier on the strength of pages describing the still-live service: the lesson is that verifying *a thing exists* is not verifying *it is still alive*, and the search to run is "<name> retired / shutdown / deprecated <year>". The failure did not stop at the dropdown entry: the rank assigned to it also made it Core Mode's automatic choice.

Mistral: mid-tier quality and a free plan requiring consent to data training — not worth paying for unpublished thesis material when five alternatives carry no such clause. *OpenRouter*: small free models with ids that change constantly, unsuited to an agent chaining ten tool calls. *Cerebras*: since August 2026 there is no card-free plan, and free context is capped at 8K — Spectra sends 32 tool definitions on top of history, which does not fit.

None of the five matches a paid frontier model on the hardest problems; GitHub Models come closest, at the cost of low request ceilings. That is why the choice stays the user's rather than imposing one service.

The paid frontier

Free tiers are enough for studying; for a hard problem they are not. The selection criterion was **GPQA Diamond** — graduate-level physics, chemistry and biology questions — because it is the only benchmark measuring what this app actually asks.

Service	Why it is here
GPT-5.6	94.6% on GPQA Diamond: the highest score available today. ~\$4/M input
Gemini 3 Pro	Over 1M context, and the same key as free Gemini. ~\$2/M
DeepSeek V4	High-tier reasoning at ~\$0.66/M: the best quality/price ratio
Grok 4.6	First on agentic tool calling and lowest hallucination rate — exactly the profile an agent chaining tools needs. ~\$2/M
Claude (Fable 5.1, Opus 5, Sonnet 5)	One key for all three; the only ones with the Python sandbox in Core Mode

Two configurations can use **the same key**: the four Claude models are one Anthropic account, and Gemini Flash and Gemini 3 Pro one AI Studio key. `chiaveCondivisaCon` points them at the same entry, so the user pastes it once. Clearing it clears it for all twins — leaving one behind would make the key reappear on the other.

The two Gemini entries forced the scorer to be generalised: they share family, endpoint and key but want **opposite** models. The preference is now declared in configuration (`/flash/` versus `/pro/`) and outweighs the rest of the score. Without it the free configuration would have picked `gemini-3-pro` — a newer generation, therefore a higher score, but **paid**: a model the free key cannot use.

The same pass surfaced a second issue: the minor version in the name is **optional**. From generation 3 Google dropped it (`gemini-3-pro` , not `gemini-3.0-pro`), and the regex requiring the dot scored the *newest* models as generic aliases — so they lost to older ones, and the automatic upgrade described in 2.4 would never have triggered.

The provider audit, and why it must be repeated

The system repairs itself on **model names** — it queries the catalogue and picks among what exists. It does not repair itself against a **dead service**: that must be checked by hand, and GitHub Models' retirement showed what not doing so costs.

September 2026 check, provider by provider:

Provider	Result
Groq	Active. <code>llama-3.3-70b-versatile</code> is no longer served as of August 2026 and <code>llama-3.1-8b-instant</code> is deprecated: removed from candidates, leaving Groq's own replacements
Google Gemini	Active
NVIDIA NIM	Removed 09/2026: alive, but without CORS headers it cannot be called from a web page
Z.AI	Active, GLM-4.7-Flash and 4.5-Flash still free
OpenAI, DeepSeek, Anthropic	Active
xAI	Active, but candidates were two generations behind: the top is <code>grok-4.6</code> since 12/08/2026

Two traps recorded so they are not repeated. First: never put a paid model among a free configuration's candidates — true for `glm-5.x` on Z.AI as it was for `gemini-3-pro` , or the configuration picks something the free key cannot use. Second: a dead provider with a high rank does not merely break a dropdown entry, it **becomes Core Mode's automatic choice**.

When a provider does not answer

A `fetch` that fails **before** receiving a response means one of three things: no network, the service does not accept direct browser calls (CORS), or the service no longer exists. The browser, for security reasons, does not let you tell which — the error is deliberately generic.

This used to stop the turn. But if the three causes cannot be told apart, the right response is the same for all three: **move to another provider** the user holds a key for, exactly as for exhausted quota. It is the behaviour that would have masked GitHub Models' retirement entirely, instead of leaving every request stuck on a dead service.

Errors that would repeat identically everywhere stay out of the fallback — a wrong key, a malformed request: trying them on every provider and then reporting a random last error would hide the real cause.

Removing a service from the list is a breaking operation, and deserves a note: whoever had it selected has that name saved in `localStorage`, and without validation would hit `PROVIDERS[undefined]` — Spectra would no longer open. `getSavedProvider()` always validates against the registry and falls back to an existing service.

2.5 The optional proxy — Spectra without a key

The API key lives in the user's browser: with no backend there is no alternative, and it is the limitation stated openly at the end of this document. Every visitor has to obtain their own free key and paste it in.

`proxy/spectra-proxy.js` is the way out, and it sits **outside** the page: a Cloudflare Worker (free plan) holding the keys as server-side secrets. When it is configured the browser carries no key at all, and anyone opening BioSpecInfo uses Spectra without entering anything.

It is not a plain forwarder:

- **The path passes through untouched.** The route is `/<provider>/<path>` and the rest is relayed as received: the proxy knows nothing about API shapes and keeps working when Spectra changes model or parameters.
- **Several keys per provider, with automatic takeover.** Each secret is a comma-separated list; on `429` (quota) or `401/403` (revoked key) it moves to the next one within the same request. Not on `400` — that is a bad request and would fail identically.
- **Streaming is never buffered:** the response body is passed straight through, otherwise answers would appear all at once at the end.
- **It is not an open relay.** Credentials arriving from the client (`Authorization`, `x-api-key`, `?key=`) are discarded and replaced: nobody can use the proxy to push a key of their own.
- **Three brakes** against abuse: an allowed-origins list, a per-IP limit and a daily cap.

On the Spectra side the graft is in one place. `GET /stato` reports which providers the proxy actually covers, so a model whose secret is missing is never offered as "no key needed"; `buildRequest` routes to the proxy **and omits authentication entirely**; `chiaveDaUsare()` returns a placeholder so the "missing key" guards do not block sending — a placeholder that never leaves the browser.

Both paths coexist: without `PROXY_URL` everything works as before, and for providers the proxy does not cover Spectra keeps using the local key.

2.6 Starting over

Everything Spectra remembers lives in `localStorage`: chats, history, API keys, persistent memory, review cards. A "clear history" button that removed only the conversations would not start over — reopening the app would restore the same state.

The delicate part is that keys exist in **two formats**: the current map (`bsi_api_keys`, one key per provider) and the single-key format of earlier versions (`bsi_api_key`). `getKeysMap()` migrates the latter into the former, so clearing only the map **brings the old key back** on the next access. Both must go — along with the resolved-model caches, which mean nothing without a key.

That is why the list of everything the application writes lives in one place, `DATI_CANCELLABILI`, split into five groups with a label and a description. The confirmation panel is built from it: it shows how many entries actually exist per group, disables the empty ones and requires a second explicit click. Only chats and keys are pre-selected; memory, review scheduling and the rest of the app's progress must be chosen deliberately, because they erase work unrelated to starting the conversation over.

Deliberately excluded: the PRO licence, the trial period and the device identity — data a user does not expect to lose by emptying a chat, and a licence must never be deleted by accident. The panel says so.

After deletion the in-process state is reset too — the list of providers the proxy covers, the resolved model per provider — otherwise it would stay valid until the page reloads, and Spectra would keep using a model chosen with a key that no longer exists.

2.7 Making free keys carry real load

Free tiers impose two different limits, and each needs a different answer.

The per-request ceiling. GitHub Models accepts 8,000 input tokens. Spectra's fixed cost is ~8,150 — 2,009 for the system prompt plus **6,128 for tool definitions alone** — so that service would not even start: it would fail before the user typed a word. The measurement came before the fix, and changed what the fix had to be.

The answer is not to trim history — it is to trim **the fixed cost**. Tools are paid on every turn; the conversation is the content. Of the two, the fixed cost must shrink first. `adattaAlBudget()` selects the tools **relevant** to the question instead of all 32, then shortens history with what remains.

Relevance is scored against a keyword map kept **outside** the tool registry: the words people use to *ask* for something are not the words a tool is *documented* with. The description of `astrofisica` talks about Wien and Stefan-Boltzmann; the user writes "what temperature is this nebula". Matching the question against descriptions alone picked `farmacocinetica` for an astrochemistry question — measured, not assumed. Matching is by stem (first five characters) because Italian inflects: "nebulosa" and "nebulose" must count as the same word.

Rate limits. A 429 on a free tier is almost always the *per-minute* limit, not an exhausted quota: waiting a few seconds clears it. The provider says how long in `Retry-After`, or in the error text ("*Please try again in 7.5s*"); when it says nothing, exponential backoff with a little jitter, so every open tab does not retry at the same instant. If the requested wait is absurd, it does not wait at all: better to say "quota exhausted" than to leave the user watching an hourglass for an hour.

Provider fallback. When a quota really is exhausted, the turn is redone on another service the user holds a key for. This is why several free keys together carry load that none of them carries alone.

Two non-negotiable constraints:

- It restarts from the **original** messages, not from half-finished history.

Turns containing tool calls are stored in the previous provider's native format and cannot be handed to another. The turn's work is lost; a correct answer is gained.

- Fallback uses **free services only**. Silently moving to a paid one would spend the user's money without their say-so; if the chosen service was paid, it still gets the first attempt.

And it moves on **only** for exhausted quota. A 401 or a malformed request would repeat identically everywhere: trying all providers and then reporting a random last error would hide the real cause.

2.8 Core Mode, and saying what is happening

The mode. One switch that changes four things at once, because none of them alone would move the needle: it switches to the most capable configuration among those with a key, raises agentic rounds from 10 to 16 (a long tool chain ran out halfway), requests maximum reasoning depth from models that expose it, and adds a system-prompt instruction to work in verified steps rather than answer fast.

The model switch is **announced in the chat**, and if the new one is paid it says so: switching silently to a metered service would spend the user's money without telling them. For the same reason it is a switch and not the default — it costs more quota, and on a paid service it costs money.

Saying what is happening. Several seconds can pass between sending and the first word of the answer: the model reasons, calls tools, waits on the network. Before, the user saw a mute pause and concluded it had frozen.

The header now carries an atomic core whose orbitals change speed and colour with state — slow and cyan at rest, fast while reasoning, amber while a tool runs — alongside a line saying the same thing in words, for anyone not watching the animation. The states hook into callbacks the agentic loop already emitted: no mechanism was added, only made visible.

Three layout defects fixed in the same pass, all found by measuring rather than looking:

- The input field shared its row with four buttons and stayed 156 px wide: the

placeholder wrapped and was **clipped**. The text now has its own row with the controls below it. The placeholder length was chosen by testing at 360 px width, not by eye.

- The title "Spectra — il copilota AI di BioSpecInfo" did not fit and truncated mid-word. Only the name remains; the rest became the status line, which now has a purpose.
- The core's orbitals, when rotating, escaped their box and overlapped the name.

2.9 Using what the provider actually offers

Three things the Anthropic API provides that were not being used. Checked against the official reference rather than recalled — two of the three carry a constraint that would not have been guessed.

Prompt caching. The fixed cost of every request is about 8,150 tokens: 2,000 of system prompt plus 6,128 of tool definitions, identical every turn. Marked for caching they are paid once and then read back at a fraction of the price.

The constraint is that caching works by *prefix*: one differing byte at the start invalidates all of it. But Spectra's prompt also carries user memory and question context, which change every message. Concatenated into one string the cache **would never have hit**. The prompt now travels in two blocks — stable and marked, volatile and not — and the marker also goes on the last tool, because composition order is `tools` → `system` → `messages`.

The Python sandbox, and why it excludes web search. `code_execution` gives the model an environment with sympy, numpy, scipy and matplotlib: numerical integration, symbolic algebra, matrix diagonalisation — things the 32 solvers do not cover because they cannot all be anticipated.

But the new-generation web search **already carries an execution environment** (it filters results by writing code), and declaring `code_execution` as well would create a second one, confusing the model. They are alternatives, not complements. The choice follows Core Mode: normally web search and page fetching, in Core Mode the sandbox — because when maximum power is requested the problem is computational, not documentary. A test verifies the two never appear together.

Page fetching. `web_fetch` joins search: if the user pastes an article link, Spectra reads it instead of merely searching for its title.

Sandbox results arrive as `bash_code_execution_tool_result` and `text_editor_code_execution_tool_result` — not a generic `code_execution_tool_result` — and must be kept in the assistant turn like the web-search blocks, or a server-paused turn cannot be resumed.

3. The 35 tools

The list is the one **measured** in `window.BSI_AI_T00LS` on the running application, not transcribed from memory: this document declared 32 when there were already 35, and three — `analizza_molecola`, `disegna_molecola`, `mostra_spettri` — did not appear at all.

Area	Tools
General computation	calcola, risolvi_equazione, analisi_dati
General chemistry	bilancia_equazione, stechiometria, massa_molecolare, converti_unita, costante_fisica
Physical chemistry	termodinamica, equilibrio_acido_base, cinetica, gas_e_soluzioni, elettrochimica
Structure & spectra	spettroscopia, quantistica_e_spettroscopia, cristallografia, analizza_molecola, disegna_molecola, mostra_spettri
Life sciences	biochimica, farmacocinetica, valuta_druglikeness
Physics	astrofisica, nucleare, statistica_inferenziale
External databases	cerca_pubchem (NIH), cerca_letteratura (PubMed)
Internal data	cerca_nel_database (9 datasets), cerca_molecola
App control	naviga_sezione (87 sections), apri_strumento (12 labs), stato_app
Memory	ricorda, ricordi
Animations	apri_animazione (6 reaction mechanisms)

Web search is not among these thirty-five, and the distinction matters. The thirty-five are executed **by the application**, in the user's browser. On providers that offer them, the agent adds two tools to the schema that are executed **by the provider** — `web_search` and `web_fetch`, with a per-turn usage cap. Counting them with the others would make them look always available, whereas they depend on the provider chosen.

3.1 Internal datasets exposed

297 synthesis reactions · 118 elements · 67 amino acids · 178 drugs · 63 pathologies · 46 retrosynthetic strategies · 36 drug interactions · 29 metabolic pathways · 29 redox potentials.

Drugs and strategies are the two figures that `tools/verifica-affermazioni.js` measures against the running application on every battery run: here they had been left at 143 and 39 while the app counted 178 and 46.

These are the datasets curated by the author for the application: the agent treats them as a primary source and is instructed to **report discrepancies** between the database and its own memory rather than silently smoothing them over.

4. Notable engineering decisions

This section documents the non-obvious choices and the reasoning behind them.

4.1 Expression engine without `eval()`

Problem. Serving arbitrary physical-chemistry computations requires an expression evaluator. `eval()` would solve it in a few lines.

Why it was rejected. The agent reads untrusted external content (PubChem results, PubMed abstracts, web pages). An injection in that content could induce the model to emit a hostile expression; with `eval()` that string would execute in the page context, **where the user's API key lives**.

Solution. Recursive-descent parser with a *closed* set of 27 functions and constants. It has no access to global scope: it can only do arithmetic. Validated against 10 escape attempts (`window.localStorage`, `constructor`, `this`, `fetch(...)`, calls to non-whitelisted functions) — all rejected, including when executed in a real browser.

4.2 Preserving reasoning blocks

On models with adaptive reasoning, `thinking` blocks are part of the assistant turn and must be **echoed back unchanged, cryptographic signature included**, on the next tool-use round. Dropping them causes the request to be rejected.

The implementation accumulates `thinking_delta` and `signature_delta` from the stream and reinserts them **first** in the reconstructed turn. Verified in integration tests: signature preserved, ordering correct.

4.3 Resuming paused turns (`pause_turn`)

Web search is a server-side tool. When its internal loop exhausts its iterations, the response ends with `stop_reason: "pause_turn"`.

The turn must be resumed by **sending the assistant turn back as-is, without adding any user message**: the server recognises the trailing `server_tool_use` block and continues from there. Appending a "continue" message would break the mechanism.

4.4 Per-model configuration, not global

The four Claude tiers do not accept the same parameters:

Model	effort	Web search	Notes
Fable 5.1	<code>xhigh</code>	yes (8)	refusal fallbacks; reasoning always on
Opus 5	<code>high</code>	yes (6)	default
Sonnet 5	<code>high</code>	yes (5)	
Haiku 4.5	unsupported	unsupported	sending them returns HTTP 400

On all recent models `temperature`, `top_p` and `budget_tokens` were removed from the API and their presence causes an error: none of the four sends them. Every parameter is therefore conditioned on the provider rather than set globally.

4.5 Multimodal input and material reading

The agent accepts images, PDFs and text files, several at once, treated as consecutive pages of a single document. Three constraints shaped the implementation:

- **Images are downscaled to 2000 px** and re-encoded as JPEG before sending.

A phone photo drops from roughly 1 MB to 440 KB: without this step a single image would approach the request ceiling and multiply token cost. 2000 px is the minimum for reading small handwriting — at 1600 px fine script became illegible.

- **The PDF page ceiling depends on the model**, it is not a constant: 600

pages for models with a 1M-token window, 100 for 200K ones. Page counting is done by reading `/Type /Page` objects from the file bytes, without libraries.

- **History stores a 160 px thumbnail**, not the image that was sent. Storing

the large one (440 KB each) filled `localStorage` within a dozen photos, and once the quota is exceeded every subsequent write fails silently.

Six quick actions turn the material into a study artefact (summary, concept map, outline, flashcards, exam questions, transcription) and two translate it. For transcriptions the agent is instructed to write `[illegible]` rather than guess: an invented chemical term is worse than a declared gap.

4.6 In-house graph renderer, no CDN

Concept maps are produced by the model as Mermaid diagrams and drawn by a renderer written into the application (~190 lines): longest-path layering from the roots with cycle breaking, barycentre reordering over three passes to reduce edge crossings, SVG output with Bézier curves.

Not loading Mermaid from a CDN follows from the app's offline-first nature: a map must be drawable without a network. Unsupported syntaxes are recognised and left as a readable code block instead of being drawn badly.

4.7 Voice command with wake word

User-activated continuous listening: the phrase "Hey Spectra" followed by a command sends it automatically. Two non-obvious details:

- the browser's speech recognition **stops on its own** after a few seconds of

silence and must be restarted, otherwise listening dies at the first pause;

- **all transcription alternatives** are examined, not just the first, and the

most common phonetic variants are accepted: Italian recognition renders "Spectra" in many different ways.

Denying microphone permission stops the loop instead of retrying forever. The word "spettroscopia", extremely frequent in this domain, was verified as non-triggering.

4.8 Guard against non-finite results

A systematic audit revealed that, given legitimate degenerate inputs (zero wavelength, zero volume, zero half-life, a transition between identical levels), eight solvers returned `ok: true` with `Infinity` or `NaN` among their fields.

This is the most insidious failure in this context: an exception is noticed, a formally valid but meaningless value is reported to the user as correct. The fix is a single check at the point where all results pass through: if a numeric field is not finite, the result becomes an explicit error naming the fields and stating the probable cause. It also covers tools added in the future.

4.9 Graceful degradation

- **Models without reliable function calling** (some free tiers): on the first tool-related error the request is retried once in text-only mode instead of failing.
- **Idle timeout (45 s)**, reset on every byte received: a long answer is never truncated, but a dead connection is reported immediately with an explicit message.
- **localStorage quota**: history is capped (30 conversations, 100 messages) with progressive pruning. Without this cap, exceeding the quota made every subsequent write fail *silently*, **including saving the API key**.
- **Classifier refusals** (`stop_reason: "refusal"`, HTTP 200): detected and surfaced, instead of appearing as an empty answer.

5. Verification

Verification was carried out at three levels, with recorded outcomes.

5.1 Numeric correctness — against reference values

Area	Case	Expected	Obtained
Balancing	<code>KMnO4 + HCl → ...</code>	2:16:2:2:8:5	✓ exact
Molecular mass	<code>K4[Fe(CN)6]</code> (nested)	368.35	368.345
Molecular mass	<code>CuSO4·5H2O</code> (hydrate)	249.68	249.677
Equilibrium	acetic acid 0.1 M, K_a $1.8 \cdot 10^{-5}$	pH 2.874	2.875
Thermodynamics	ΔG (N_2O_4/NO_2) at 298 K	+4.79 kJ/mol	✓
Kinetics	E_a from $k(300\text{ K})$, $k(320\text{ K})$	55.3 kJ/mol	55.33
Real gases	van der Waals CO_2	−10 % deviation	−10.13 %
Spectroscopy	bromine M+2	97.3 %	✓
Biochemistry	glycylglycine	132.12 Da	✓
Pharmacokinetics	$t_{1/2}$ from V_d 50 L, Cl 5 L/h	6.93 h	✓

Area	Case	Expected	Obtained
Astrophysics	solar emission peak (Wien)	501.5 nm	501.52
Astrophysics	Earth escape velocity	11.19 km/s	11.186
Nuclear	⁵⁶ Fe binding energy	8.79 MeV/nucleon	8.790
Statistics	critical t, 95 %, df = 4	2.7764	✓
Crystallography	copper density (fcc)	8.96 g/cm ³	8.935

P-values are not tabulated: they are computed with the regularised incomplete beta function (Lentz continued fraction) and the incomplete gamma function.

5.2 Robustness — cases that must fail

Verified as **rejected**: unbalanceable equations, formulas with unbalanced parentheses or non-existent elements, species absent from the equation, invalid database types, and the 10 injection attempts against the expression engine.

5.3 Integration — real browser

Automated tests with headless Chromium across all 14 application pages: no JavaScript errors, no missing resources. All 87 sections of `index.html` and the 18 tabs of the Astrochemistry module were traversed one by one. Request bodies were inspected for the four Claude tiers, and a full cycle with web search, turn suspension and resumption was simulated.

6. Quantitative summary

Metric	Value
Component size	6,254 lines
Tools	32
Model configurations	10 (3 free)
Scientific areas covered	13
Records in exposed internal datasets	over 800
Max agentic loop rounds	10
History sent to the model	40 turns
Runtime dependencies	none

7. Stated limitations

Documented for technical honesty; these are deliberate choices, not omissions.

- **The API key lives in the browser** — unless the proxy is deployed. In

the default configuration the key never leaves the device, but it is accessible to anyone with access to that browser, and every user must obtain their own. The Worker in section 2.5 removes this limitation by moving keys to a server, at the cost of one component to maintain outside the page: a distribution choice, not a requirement.

- **Network tools depend on service availability.** PubChem and PubMed are queried directly from the browser; when unreachable, the tool returns an explicit error and the agent is instructed to say so rather than substituting an estimate.
- **Solvers apply simplified models** where the literature offers more refined ones (e.g. Woodward-Fieser for UV, or ATP yields using average P/O ratios). Each tool states the method used in its own result.
- **Verification is functional and numeric**, not formal: there is no proof of solver correctness, but a set of reference cases drawn from the teaching literature.

Companion document to the BioSpecInfo technical dossier. See also: [01-Software-Architecture-Document.md](#), [02-Verification-Validation-Report.md](#), [03-Security-Privacy-Compliance.md](#).

Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	bsi-v172
Commit	944718849558dc9b3540173914034f2cf1abe280
Generated (UTC)	2026-09-23T13:16:24.213Z
Machine-readable format	<code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	<code>RDKit_minimal.js</code>	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	<code>RDKit_minimal.wasm</code>	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	<code>three.min.js</code>	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	<code>three_bloom.js</code>	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	<code>gltf_loader.js</code>	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	<code>3Dmol-min.js</code>	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).
SmilesDrawer _{Daniel Probst}	<code>smiles-drawer.min.js</code>	MIT	246 kB	2D structure drawing from SMILES.
sql.js (JS glue) _{sql.js / SQLite}	<code>lib/sql-wasm.js</code>	MIT	45 kB	In-browser SQLite database for the data exercises.

Component	Version/File	Licence	Size	Role
SQLite (WebAssembly) _{SQLite Consortium}	lib/sql-wasm.wasm	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	lib/dimuon.js	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
RDKit_minimal.js	58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9
RDKit_minimal.wasm	e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
gltf_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d

2. Code written for this project

File	Size	Role
index.html	4537 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	32 kB	IR/NMR spectral prediction engine over the molecular graph.
sw.js	5 kB	Service Worker: offline operation and network strategy.
rdkit_lab.html	257 kB	Cheminformatics laboratory.
astro.html	2666 kB	Astrochemistry module.
chimorga.html	194 kB	Organic chemistry module.
proxy/spectra-proxy.js	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
index.html	8145b05ac943c52de5d51f92f378dc924fa436342a5dcb249db9b4b1e3d38e35
bsi-ai-hub.js	985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476

File	SHA-256
bsi-spettri.js	6f6d13824b593c92edfdf51cd976de4cb72479c9c6586c5158c56dab615a779b
sw.js	6652fbc795f983e6cf4d2b55187206ed6100f3bad72e9fd542b05cc4eb747193
rdkit_lab.html	e80356f161d82e35906dc40df5ec70158e6247d473682398d7b1c8a718fe175f
astro.html	53e8d4d29af4bb9cfdb63d974924737e606e3141bd4ce61fbab1bbebe0eed30d
chimorga.html	047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d
proxy/spectra-proxy.js	808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
playwright-core	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

The application's licence is distinct from those of its components. The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see [LICENSE](#)): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

Generated on 2026-09-23T13:16:24.213Z.

Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	bsi-v172
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

How to read this matrix. Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	Structural data for drugs must be consistent with the declared molecular weight	FARM_DATA in <code>index.html</code>	<code>tools/verifica-farmaci.js</code> — RDKit ↔ literature comparison over 178 entries
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	<code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit	<code>test_spettri</code> — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	<code>bandeDaGruppi()</code> , additive construction	<code>test_spettri</code> §1 — acetylsalicylic acid: ester and acid
SCI-04	Spectra must respect the conventions of their technique	<code>svgIR()</code> , <code>makeNMRsvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code>	<code>test_assi</code> (SVG), <code>test_assi_canvas</code> (canvas)
SCI-05	A computed spectrum must be reproducible	every <code>Math.random()</code> removed from the traces	<code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison
SCI-06	The height of ¹ H peaks must represent the integration, or not be asserted at all	<code>makeNMRsvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height	<code>test_assi</code> §4
SCI-07	Conjugation must act on the individual carbonyl	SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code>	<code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690

ID	Requirement	Implementation	Verification bench
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	<code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate	<code>test_costanti</code> — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; <code>docs/06 §3.5</code>	<code>verifica_guida</code> — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	<code>ssimm</code> section in <code>index.html</code>	<code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	<code>scroma</code> section: $\sigma = t_{R}/\sqrt{N}$, R_s from distance and widths	<code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	<code>staratura</code> section: ordinary least squares, F-test line↔parabola, deleted residual	Verified on the three example sets included: linear ($F = 0.3$), non-linear ($F = 186.6 > 7.71$), with an outlier (1 detected)

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	<code>modelloSuggestitoDaErrore()</code> , <code>bsi_modelli_ko</code> blacklist	<code>test_404</code> — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	<code>bsi_prov_ko</code> , 24 h TTL, <code>providerUtilizzabili()</code>	<code>test_ko</code> (35), <code>browser_ko</code> (37)
AI-03	A temporary overload must produce a retry, not an error	<code>erroreTemporaneo()</code> , <code>TENTATIVI_TEMPORANEO</code>	<code>test_503</code> — 40 checks
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	<code>ATTESA_TROPPO_LUNGA_MS</code>	<code>test_attesalunga</code> — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	<code>tettoDaErrore()</code> , <code>MARGINE_TETTO</code>	<code>test_tetto</code> — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in <code>appendAgentTurn()</code>	<code>test_firma</code> — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	<code>_turnoInCorso</code> in <code>send()</code>	<code>test_doppioinvio</code> — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	<code>provaTuttiIFornitori()</code>	<code>browser_prova</code> — 29 checks

ID	Requirement	Implementation	Verification bench
AI-09	Paid providers must not be offered as free ones	separation by <code>PROVIDERS[id].free</code>	<code>browser_prova</code> §D2
AI-10	A documented route must exist to remove CORS and rate limits	<code>proxy/spectra-proxy.js</code> + in-app guide	<code>test_guidaproxy</code> (12), <code>browser_proxy</code> (15), <code>browser_proxyui</code> (17)

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	<code>audit_stabilita</code> §1 — 87 sections × 5 rounds
STA-02	Exhausting <code>localStorage</code> must not make the app unusable	guarded writes; persistent warning in the File Manager	<code>audit_quota</code> (10), <code>test_filemanager</code> (15)
STA-03	No rejected promise may remain unhandled	systematic <code>.catch()</code>	<code>audit_promesse</code> — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	<code>loadJSON()</code> with fallback	<code>audit_stabilita</code> §4
STA-05	A slow network must not be treated as an absent one	<code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>	<code>test_sw</code> §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	<code>BSI_SW_UPDATED</code> message, decision taken page-side	<code>test_sw</code> §4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	<code>test_visore3d</code> — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	<code>dbStore()</code> over a promise; persistent warning	<code>test_filemanager</code> §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	<code>audit_stabilita</code> §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	<code>bsiNitido()</code> on the canvas contexts	<code>audit_grafici</code> — 40 canvases
UI-03	The app must work at 390 px width	auto-fit grids, no fixed column	<code>audit_mobile</code> — the document's <code>scrollWidth</code> across all 87 sections at 390 px; <code>audit_stabilita</code> for errors in a phone viewport

ID	Requirement	Implementation	Verification bench
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	test_aggiorna — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	bsiCancellaDati() by group	browser_reset — 24 checks

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in localStorage or in the Worker's secrets	tools/verifica-sicurezza.js — 235 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	audit_rete — direct verification : a canary value seeded into 71 stores of the user's data, the application used across 87 sections on 6 pages, and the URL, headers and body of every request inspected. Plus verifica-sicurezza on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in file_manager.html	tools/verifica-sicurezza.js
SEC-04	No conflict marker may reach publication	—	verifica_guida §12 — 58 text files

SEC-03 — declared limitation. GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal: the only effective remedy is to change it.

6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across 13 pages and 87 sections (<code>tools/verifica-accessibilita.js</code> , 33,642 text elements: it was 19,751 before text over gradients entered the measurement). Of the 1,069 contrast defects that emerged, none remain: 0 measured over those same 87 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are zero : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (<code>docs/02</code> §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	not representable : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4
PERF-02	Code coverage of the verification benches	partially measured : 49.79 % of statements over the widest path (<code>audit_copertura</code>). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	not verified — the battery runs on Chromium only; see §8

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01...13)	12	12	100 %
AI agent (AI-01...10)	10	10	100 %
Stability (STA-01...08)	8	8	100 %
Interface (UI-01...05)	5	5	100 %
Security (SEC-01...04)	4	4	100 %
Automated total	39	39	100 %
Not automated (§6)	5	0	0 %
Declared total	44	39	89 %

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

Why two totals. Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **39 requirements verified out of 44 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: 49.79 % of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (<code>c8</code> , <code>istanbul</code>), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers 403 to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (<code>SCI-05</code>): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

Document updated to version `bsi-v172`.

Release Conformance Statement — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version	bsi-v172
Author and release owner	Samuele Pio Provenzano
Repository	github.com/samupropiol-ship-it/BioSpecInfo-v11
Distribution	GitHub Pages — samupropiol-ship-it.github.io/BioSpecInfo-v11/
Nature of the software	Client-side Progressive Web App, no backend

1. Subject of this statement

The undersigned declares that version `bsi-v172` of BioSpecInfo has been put through the verification procedure described in `02-Verification-Validation-Report.md` and that the outcomes are those reported, without selection, in `../evidence/RAPPORTO-VERIFICA.md` — a document generated automatically by `tools/genera-evidenza.js`, which transcribes the benches' output with no manual intervention.

This statement does **not** attest the absence of defects. It attests that:

1. an automated verification procedure exists and can be reproduced by third parties;
 1. the outcome reported is the one actually produced, failures included;
 2. the known limitations are declared explicitly and not presented as features.
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2. Intended use and limitations

2.1 Declared purpose

BioSpecInfo is a **teaching and study-support** tool for chemistry, biochemistry, pharmacology and astrochemistry. It is designed for university students and for exam preparation.

2.2 What the software is NOT intended for

These exclusions are an integral part of this statement.

Domain	Exclusion
Clinical use	It is not a medical device within the meaning of Regulation (EU) 2017/745. It does not support diagnosis, prescription, dosing or therapeutic decisions. The pharmacological data are for teaching purposes only, and the application says so at the head of the section.
Certified analytical use	The spectra are predictions from functional-group tables, not measurements nor quantum-mechanical simulations. They do not replace an experimental database (SDBS, NIST) nor an instrumental determination, and cannot be used for the formal identification of substances.
Regulated GxP environment	It is not validated for use in a GLP/GMP environment. The alignment with GAMP 5 described in <code>docs/03</code> is methodological, not a formal qualification.
Access control	The File Manager protects its contents with a client-side deterrent. On a static site there is no effective access control: whoever reads the source bypasses it. It must not be used for confidential material.

3. Summary of verification outcomes

The full outcomes, with each bench's textual output, the execution environment, the commit and the SHA-256 digests, are in `../evidence/RAPPORT0-VERIFICA.md`.

Family	What it demonstrates
Scientific data	The chemical data are verified against an independent source
AI agent	The assistant stays usable when the external provider fails
Stability	The app withstands long sessions, exhausted storage, degraded networks
Interface	The panels respond as documented
Documentation/code consistency	What the documentation promises exists in the code

4. Known deviations at the date of release

Listed in full. None has been removed from verification in order to make it pass.

#	Deviation	Impact	Treatment
D-01	2 entries (it was 6) that are not single molecules: Ivermectin, a mixture of B1a/B1b homologues, and Coartem, a combination of two active ingredients	For these two no 2D/3D representation and no spectral prediction is available	Entries left without SMILES : the pharmacological data remain, the structure is not shown. For Digoxin, Vincristine and Tacrolimus (topical and systemic) the structure was taken from ChEMBL and passes the comparison with the declared molecular weight: those four left the registry. The two that remain have no structure to show, not a missing one.

#	Deviation	Impact	Treatment
D-02	19 entries with no SMILES by their very nature (monoclonal antibodies, peptides)	None: for those molecules SMILES notation is not the appropriate representation	Not a substantive deviation; listed for completeness
D-03	SEC-02 verified only by proxy — remedied: see S-10	—	<code>audit_rete</code> observes the traffic during real use, with a canary value seeded into the user's own data
D-04	The File Manager password is present in the git history predating its removal	The deterrent is known to anyone consulting the history	Documented; the effective remedy is to change the password
D-06	Code coverage measured but partial: 49.79 % of statements, not branches	It is the coverage of the widest path a bench walks (87 sections plus the other pages), not of the whole battery; and an <code>if</code> entered from one side only counts as covered	<code>audit_copertura</code> measures it with Chromium's profiler, with no build and without rewriting the source. The value is recorded: if it falls, the battery fails
D-07	Verification on Chromium only	Firefox and WebKit are tested by hand	Declared in <code>docs/08</code> §8. In the verification environment the reason is checkable: the CDN from which <code>playwright-core</code> downloads the other engines answers 403 to the network policy, so Firefox and WebKit cannot be installed there
D-08	English translation covering 10 of the 16 documents	A non-Italian-speaking assessor reads 10 documents out of 16	Declared here; the remaining ones are available in Italian. The translated set covers the whole due-diligence path: dossier, architecture, V&V, security, licences, AI agent, SBOM, traceability, this statement and the test documentation
D-09	Full WCAG 2.1 AA conformance not verifiable entirely by automation	What stays outside is text inside SVGs (8,888 elements), text over a real background image (108) and everything requiring human judgement. Text over a gradient has entered the measurement: 591 elements, judged against the worst stop of the gradient	The skipped elements are counted and reported on every run

Contrast, from 1,069 defects to zero — how it surfaced and how it was closed

Up to version bsi-v168 this document declared WCAG conformance "verified by sampling" and the bench reported **0 defects**. Those were two statements at once true and meaningless: the bench inspected only the **visible** elements, and index.html shows one section at a time. Out of 19,751 text elements it was looking at **41**.

Walking all 87 sections, **1,069** contrast defects surfaced. They had not appeared: they had always been there.

Today they are **zero**, measured over those same 87 sections with the same bench. Not by wholesale substitution — that road, attempted once, introduced 997 new defects — but by going back to the **common cause** every time and re-measuring after each change. The **40 unlabelled fields** have likewise fallen to **zero**.

The gravest causes were not wrong colours but **theme tokens whose value equalled the background**:

Structural defect	Measured effect
--g800 is #0d1522 , identical to --bg — and the rule a { color: var(--g800) } applied it to every link	Every link without a colour of its own was at contrast 1:1 : invisible
--g700 is #16263d , nearly identical to --white (#16273e)	Text at 1.01:1
Buttons that changed their background without changing their text colour	The selected state (or the deselected one, depending on the panel) became unreadable: 1.39:1
lightenColor(hex,pct){ return hex; }	A function called "lighten" that returned the colour unchanged : an unfinished stub, in all likelihood written for this very problem
The quiz answer panel: light background, light text	The student could not read the correct answer

The structural remedy was an explicit token — --testo-forte — for prominent text on a dark surface, applied only to the uses as a *text colour*: --g800 and --g700 stay where they serve as background or border.

Cause	Defects	Remedy
A dark-theme safety net with higher specificity than the components: text meant for the white cards was forced to #d4dce6 , contrast 1.38:1	159	Rewritten with :where() (zero specificity): it protects text that has no colour of its own and stops fighting text that has one
A per-category accent used as text on a dark header, with #0d1522 (near-black) as fallback: 1.13:1	~400	bsiAccentoLeggibile() preserves the hue and raises the lightness just enough. Applied at the 14 points where the accent is chosen — not in the data, so it holds for future entries too

Cause	Defects	Remedy
Two hues used only on dark backgrounds, just under threshold (3.75:1 and 3.04:1)	139	Replaced after verifying that they never appeared on a light background
Table zebra striping white/near-black with always-light text: the white rows were unreadable	57	Two dark tones, consistent with the theme
Periodic-table cells just under threshold (4.46:1 and 3.88:1)	43	Text at #e4eff4, verified across all nine categories
"#var(--text)" — one # too many makes the colour invalid: in SVG it falls back to black, on canvas the assignment is ignored and the previous colour persists	8 places	Replaced with the real value

The direction of the correction depends on the surface, not on the colour. The first attempt always lightened: it removed thirty defects on the dark header and added eighteen on the light formula box that uses the same accent. The function now darkens on light backgrounds and lightens on dark ones.

The final tail: 80 → 0. The last eighty were scattered over 33 distinct causes and looked like finishing work. They were not: three more structural causes lay underneath.

Final cause	Remedy
The "automatic contrast corrector" was producing defects instead of removing them. It used <i>perceived</i> brightness instead of WCAG luminance, so its thresholds corresponded to no real ratio; it ignored alpha , so <code>rgba(255,180,84,.08)</code> over a dark surface looked to it like a "light background" and it darkened the text to #16273e — creating a defect at 1.21:1 on top of a source that was already correct; and it looked only at elements carrying a background of their own , that is, never the commonest case	Rewritten: it walks the ancestors compositing alpha to obtain the effective background, measures the true WCAG ratio, and delegates the correction to <code>bsiAccentoLeggibile()</code>
--g900 is #0d1522 — the same as --bg , exactly like --g800: the legacy of a greyscale born for the <i>light</i> theme, where --g900 was "the darkest text". With the theme inverted, that role no longer exists	The uses as a text colour (30 rules today) move to --testo-forte; the 16 uses as a background stay where they are
A colour chosen from data used as a background with the text hard-coded to #fff: it works while the hue is dark and stops as soon as someone adds #e65100	<code>bsiEtichettaLeggibile()</code> — the dual of the previous function: it picks the text by looking at the background and, if neither extreme suffices, darkens the background while preserving the hue , because a label must stay recognisable by colour

The steps as measured with the official bench, not estimated: **80 → 27 → 26 → 10 → 4 → 0**. Each step is a full re-measurement over the 87 sections after a group of changes; none of the figures above is attributed to a single cause, because the causes were not

measured one by one.

The number remains **recorded as a baseline** in `evidence/accessibilita-riferimento.json`. Zero is not an achievement to file away: it is the value the bench defends. If a future change reintroduces even one defect, the battery **fails**.

What stays outside automation is what automation cannot judge: text inside SVGs and text over gradient backgrounds (D-09). Those elements are **counted** and reported on every run, not passed over in silence.

4-bis. Deviations remedied in this version

They are listed because they concerned **the verification apparatus itself**, and a conformance statement that says nothing about the defects of its own evidence is not worth the paper it is written on.

#	Deviation remedied	Why it mattered
S-01	32 of the 39 benches were not in the repository: they lived in the working area of the session in which they had been written	It made false the sentence this whole document rests on — "anyone can re-run them". Whoever cloned the repository found 7 out of 39. They now live in <code>tools/banchi/</code>
S-02	A missing bench was counted but did not make the battery fail	A battery missing <i>all</i> its benches would have declared "CONFORMANT" with zero failures. A missing bench now makes the outcome NON-CONFORMANT
S-03	<code>package.json</code> and <code>package-lock.json</code> were excluded from the repository	<code>npm install</code> , the documented command for reproducing the evidence, installed nothing: the battery failed at the first bench with <i>Cannot find module</i>
S-04	The accessibility check inspected only the visible section	On <code>index.html</code> that meant one section out of 87. The "0 defects" held for the Dashboard, not for the application
S-05	<code>SCI-10</code> was defined twice in the matrix, and the coverage totals were wrong (36 declared out of 37 defined)	The first number an assessor reads was wrong. A bench now verifies it
S-06	50 broken links out of 127 in the delivery packages	Documentation handed to third parties with references that led nowhere
S-07	15 attachable PDFs stuck at documents 00-05 and at an earlier version	An obsolete attachment states false things with an air of authority
S-08	The numeric-claims bench looked at the Italian documents only	The English set drifted undisturbed: <code>docs/en/05</code> declared "84 sections" while the app has 87, and stayed stuck at version <code>bsi-v146</code> while the code was at 171. The English set is now compared against the measurement just like the Italian one, and a disagreement between the two languages fails the bench

#	Deviation remedied	Why it mattered
S-10	SEC-02 — "no personal data leaves the device" — was verified by checking the two <i>mechanisms</i> of exit, not the exit	A check on mechanisms says "I cannot see how it would leave", which is not "it did not leave". <code>audit_rete</code> seeds an unrepeatable value into 71 stores of the user's data, then uses the application and inspects the URL, headers and body of every request. Proved in reverse with <code>BSI_PROVA_FUGA=1</code> , which causes a leak on purpose: the bench must fail, and it does
S-11	The deviations registry had no converse guard: an entry declared that no longer deviates simply stayed	An exemption nobody revokes is a permission left switched on, and tomorrow it would silently cover a wrong structure put in its place. The bench now fails for that too — verified by putting a remedied entry back
S-09	The "automatic contrast corrector" created the defects it was meant to remove	It ignored alpha: a <code>rgba(255,180,84,.08)</code> background over a dark surface looked light to it, and it darkened with <code>!important</code> a text that was already correct in the source. A remedy tool that worsens the thing it remedies is the hardest kind of defect to see, because it presents itself as the solution

5. Acceptance criteria applied

A version is published only if **all** the following criteria are satisfied.

- ☑ No non-network JavaScript error on any page
- ☑ `tools/verifica-farmaci.js` — no defect; every deviation recorded
- ☑ `verifica_guida` — no promise in the documentation without a counterpart in the code
- ☑ No conflict marker in the tracked files
- ☑ Service Worker cache version bumped and aligned with `BSI_APP_VERSION`
- ☑ Known deviations listed in §4 of this document
- ☑ Evidence report regenerated against the published commit
- ☑ **No missing bench** — a bench that is not there is not a bench that passed
- ☑ `tools/genera-pacchetti.js` closed with **0 broken links**
- ☑ `tools/genera-pdf.js` re-run: the attachments describe this version

6. Change management

Aspect	Procedure
Traceability	Every change goes through a pull request describing the defect, its cause and its verification
Versioning	The application version coincides with the Service Worker cache version (<code>bsi-vNNN</code>), incremented at every publication
Distribution	Automatic from <code>main</code> via GitHub Pages; the Service Worker signals the new version to open sessions without interrupting them
Regression	The full battery is re-run before every merge into <code>main</code>
Reversibility	Every version corresponds to a commit; rollback is a <code>git revert</code>

7. Intellectual property and licences

All third-party components are distributed under **permissive** licences (MIT, BSD-3-Clause, CC0, Public Domain), mutually compatible and free of strong copyleft constraints. The full bill of materials, with digests and residual obligations, is in [07-SBOM.md](#) and in CycloneDX format at [../evidence/sbom.cdx.json](#).

The code written for this project is the author's own work.

8. Data handling

The application is **local-first**: there is no backend, no personal data is collected and no persistent identifier is present. Data generated by the user (notes, chats, API keys, uploaded files) stay in the device's `localStorage` and IndexedDB and can be deleted selectively from the interface itself.

Network calls are optional and go to documented public services (§4 of [07-SBOM.md](#)). If the user configures the proxy, the API keys reside on a Worker they own and not on the device.

Details in [03-Security-Privacy-Compliance.md](#).

9. Signature

This document is drawn up by the author of the software, who takes responsibility for it. It does not constitute certification by a third-party body, and is not presented as such.

Anyone can verify what is declared here by re-running the procedure in §5 of [../evidence/RAPPORT0-VERIFICA.md](#) against the commit indicated, and comparing the SHA-256 digests of the files.

Samuele Pio Provenzano _Version `bsi-v172`._

Test Documentation — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v172
Purpose	Describe how the tests are organised, how to run them, what they cover and where they leave gaps.

1. Kinds of test present

Kind	Present	Notes
End-to-end (E2E)	✓	Playwright on headless Chromium against a local HTTP server. It is the prevailing form.
Data verification	✓	Scientific data compared against an independent source
Stability tests	✓	Long sessions, exhausted storage, degraded network
Documentation/code consistency	✓	Verifies that what the documentation promises really exists
Isolated unit tests	✗	See §6
Instrumented code coverage	✓	<code>audit_copertura</code> , with Chromium's profiler: no build, no source rewritten. See §6

Why E2E and not unit tests

The application is made of classic scripts operating directly on the DOM, with no modules and no dependency injection: there are no units that could be isolated without a deep rewrite. The defects that actually occurred — a blurred canvas, a button unreachable because its container is collapsed, an AI provider that stops answering — **would not have surfaced** from a unit test anyway: they live in the integration between code, DOM and browser.

The choice is therefore reasoned, not endured. Its limits remain, and are declared in §6.

2. How to run the tests

Requirements

```
npm install          # playwright-core only
python3 -m http.server 8899 & # RDKit WASM requires an HTTP context
```

Without a server every test fails: opening the files over `file://` prevents the WebAssembly modules from loading.

The whole battery, with a report

```
node tools/genera-evidenza.js
```

It runs every bench, transcribes its full output and produces `docs/evidence/RAPPORTO-VERIFICA.md` with environment, commit and SHA-256 digests. It exits with code `0` only if no bench fails.

```
node tools/genera-evidenza.js --veloce # skips the benches that need a browser
```

The `--veloce` option is useful for a quick check but **is not sufficient for an attestation**: the benches it skips are precisely the ones measuring real behaviour.

A single bench

```
node tools/verifica-farmaci.js # a stand-alone tool
node tools/verifica-farmaci.js --json # machine-readable output

cd tools/banchi && node test_spettri.js # one E2E bench
```

The E2E benches live in `tools/banchi/`, versioned like the rest of the repository; the `BSI_BANCHI` variable allows another folder to be pointed at.

It has not always been so, and that is worth saying. Up to version `bsi-v168` the default folder was the working area of the session in which the benches had been written: **32 of the 39 benches were not in the repository**. Whoever cloned it and ran the documented command found seven, and received no warning — because a missing bench was counted but did not make the battery fail, which closed with "CONFORMANT" and zero failures.

Both things have been corrected: the benches are versioned, and **a missing bench makes the outcome NON-CONFORMANT**. A bench that is not there is not a bench that passed.

3. Composition of the battery

43 benches, grouped by what they demonstrate.

3.1 Scientific data

Bench	Verifies	Checks
verifica-farmaci	Structure ↔ molecular weight, duplicates, recorded deviations	178 entries

Bench	Verifies	Checks
test_spettri	Functional-group recognition on reference molecules	36
test_spettri_ui	Grafting of the engine into the app's spectrum-drawing points	16
test_assi	Axis conventions in the SVG spectra	13
test_assi_canvas	Axis conventions in the canvas spectra	6
test_costanti	Physical-constant lookup and refusal of ambiguities	45
audit_dati	Consistency of the tabulated data	29
test_simmetria	Point groups, normal modes, IR/Raman selection rules	26

3.2 AI agent

Bench	Condition simulated
test_404	The provider has retired the model
test_503	Temporary overload
test_ko · browser_ko	Provider unreachable from the browser
test_firma	Opaque reasoning signatures to be returned
test_tetto	Token ceiling learned from the error
test_attesa · test_attesalunga	Timeouts and imposed waits
test_doppioinvio	Two overlapping submissions
browser_prova	Reachability measurement without a key
test_nucleo · browser_prov	Agentic loop and provider selection
caccia_ai	Malformed answers, truncated stream, 502 with HTML, Stop during the answer, panel closed halfway, hostile text

3.3 Stability

Bench	What it puts to the test
audit_stabilita	87 sections opened 5 times, full storage, corrupted data, bursts of clicks
audit_promesse	Rejected and unhandled promises, with a healthy network and a dead one
audit_quota	localStorage.setItem forced to fail on 10 pages
test_sw	Offline, degraded network, update while working
test_filemanager	Store unavailable, space exhausted, failed writes
test_visore3d	WebGL context consumption while browsing molecules

3.4 Interface

browser_reset · browser_proxy · browser_proxyui · browser_rdkit · browser_lab · browser_frontiera · test_aggiorna · test_guidaproxy

3.5 Security and accessibility

Bench	Verifies
<code>verifica-sicurezza</code>	API keys in tracked files, clear-text passwords, secrets in <code>wrangler.toml</code> , telemetry, scripts from external domains
<code>verifica-accessibilita</code>	WCAG AA contrast, accessible names, field labels, alternative text, heading hierarchy, <code>lang</code> attribute — across 13 pages
<code>audit_mobile</code>	That at 390 px the page does not scroll horizontally, across all 87 sections
<code>audit_rete</code>	That no user data leaves the device : a canary value seeded into 71 stores, the app used across 6 pages and 87 sections, the URL, headers and body of every request inspected
<code>audit_copertura</code>	How many bytes of JavaScript are actually executed while walking the application: 49.79 %, recorded and defended

The obstacle belonged to the tool, not to the problem. For three versions code coverage was declared unmeasurable, with this reason: "instrumenting would require introducing a build, which the application does not have". That was true of `c8` and `istanbul`, which rewrite the source before running it. But **Chromium collects it by itself**, inside the engine, without touching a byte of the file served:

`Profiler.startPreciseCoverage`, which Playwright exposes as `page.coverage.startJSCoverage()`. No build was needed: what was needed was noticing that the constraint belonged to the tools chosen.

The first result was 100 % on every file, including 3.6 MB of `index.html`. That was the bench measuring nothing, in its most insidious form: a perfect number. V8 emits, for each function, an outer range with the number of entries, and inside it ranges with `count: 0` for what was not executed. Summing the positive ranges sums the whole file. One does the opposite: start from the total and subtract the union of the empty ranges.

The real number is **49.79 %**, and it holds the same pact as the accessibility debt: recorded, and the bench fails if it falls. An absolute threshold ("80 %") would be an invented figure; "it must not get worse" is a promise that can be kept.

Why a canary and not a list of requests. SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit: empty telemetry variable, no script from an external domain. That is real evidence but indirect: it says "I cannot see how it would leave", which is not "it did not leave". Looking instead at the list of hosts contacted would be just as weak, because it concludes from absence of evidence.

The canary reverses the burden. An unrepeatable string is written into the user's own data — notes, chats, API keys, settings, progress — then the application is **used**, and every outgoing request is opened and read. If the string appears nowhere, it is because it really did not leave.

```
BSI_PROVA_FUGA=1 node tools/banchi/audit_rete.js # must FAIL
```

The bench carries its own proof of working: with that variable it causes a leak on purpose, towards a **declared** host — contacting it is legitimate, sending it the user's data is not. If it does not fail then, the instrument is the thing that is broken.

Why a bench on horizontal overflow. UI-03 says "the app must work at 390 px", and the matrix gave it as verified by `audit_stabilita`, which in a phone viewport looks at **JavaScript errors**: a page can have not a single one and still run a hundred and sixty-three pixels off the screen. It was a requirement declared covered and not measured. The bench measures the document's `scrollWidth` — exactly what makes the scrollbar appear — and not any element wider than the screen, which inside a container built to scroll is correct and would fill the output with noise until nobody reads it any more.

What the accessibility check does not cover, and it must be said: text inside SVGs (the colour comes from `fill`, the background is a drawn shape) and text on gradient backgrounds (they do not have a colour). The skipped elements are counted and reported, not hidden.

One section at a time. The inspection skips elements that are not visible — and rightly so: a hidden element has no contrast to measure. But `index.html` alternates 87 sections and shows only one: out of **19,751** text elements the bench was looking at **41**, and printing "0 defects". It was not a false result, it was a result on a sample nobody had declared. The sections are now opened one by one, and **the number of sections traversed is printed**.

That count earned its keep on the very first run: the first attempt traversed **zero** — Playwright's click waits for the element to be visible, and the buttons sit inside collapsed navigation groups — and the bench said so ("87 sections present, none traversed") instead of printing another reassuring zero.

3.5-bis The accessibility debt, and the pact that it must not grow

Walking every section, **1,069** contrast defects surfaced. Today they are **zero** — the list of causes and remedies is in `docs/09 §4` — and unlabelled fields have fallen from 40 to **zero**. The mechanism described below remains, and matters more than the number it guards. Demanding zero *from the outset* would have left only two roads, both bad: a verification red for ever, or one loosened until it turned green again. The third road is to **declare the number** and forbid it from growing, whatever it is. That is how 1,069 became zero: every step recorded, none of them reversible in silence.

File	Role
<code>docs/evidence/accessibilita-riferimento.json</code>	The recorded count

Situation	Bench outcome
The count rises	✗ FAILED — it is a regression
The count falls	✓ passes, and asks for the baseline to be updated
The count is unchanged	✓ passes silently: debt declared, not grown — today that value is 0 , and the bench defends it

```
node tools/verifica-accessibilita.js --aggiorna-riferimento
```

To be used **only after reducing** the defects, never to turn a regression green. Even at zero this is not full WCAG conformance: text inside SVGs and text over gradients stay outside, which automation cannot judge and which the bench **counts and reports** on every run (`docs/09 D-09`). It is the honest measure of what can be measured, with the guarantee that it will not silently get worse.

3.6 Consistency

Bench	Verifies
<code>verifica_guida</code>	That the documentation's promises exist in the code; absence of conflict markers across 58 files
<code>verifica-documenti</code>	Internal links, version alignment, index consistency, justification of the deviations, internal consistency of the traceability matrix (no duplicated identifiers, coverage totals equal to the identifiers actually defined)
<code>genera-pacchetti</code>	Rewrites the references of the extracted documents and verifies that no link is left broken in the three delivery packages
<code>genera-pdf</code>	Converts the documents to PDF and fails if a file falls below the plausibility threshold (empty conversion)
<code>verifica-affermazioni</code>	Compares every number declared in the documents — sections, drugs, diseases, tumours, strategies, modules — with the one measured in the running application , in Italian <i>and</i> in English

Why numbers must be measured, not remembered. The technical dossier opens with figures: "63 diseases (23 tumours)", "46 strategies", "25 modules". They are the first thing an assessor reads, and they are enough for everything else to be believed or doubted. Nobody was checking them: written once, they stayed put while the application changed. And indeed `chimorga.html` declared **"25 modules"** at the top and **"17 modules"** further down — two statements about the same object, in the same file, one of them false.

A lesson paid for while writing the bench. I had already concluded that "63 diseases" was an exaggeration: in the source I had found a `DISEASES` array with 12 entries, and I was about to "correct" the document. Measuring in the DOM I saw that the menu really does list 63 — the array I had found was a different, smaller one. **A count on the source is not a measurement: it is a clue.** That is why the bench opens the page, opens the section, and counts.

It also checks that the documents agree **with one another**: two documents saying different things about the same object are a defect even when one of them is right. The English translation counts as one of those documents — `docs/en/05` had been left at "84 sections" while the app had 87, and had been stuck at version `bsi-v146` for three releases — and no bench saw it, because none looked at the English set.

Why a check on the matrix. The matrix declared `SCI-10` twice — once as a requirement verified at 100 %, once as an explicitly **unverified** one — and the summary table added up to 36 requirements while 37 were defined in the document. Both errors sat in the number an assessor reads first. The bench now counts the identifiers actually present and fails if the totals written do not match: the proof that the check really measures something is that, with the two errors reintroduced one at a time, it reports both.

4. Notable test cases

Some benches deserve a mention because they verify properties a functional test does not observe.

Case	Method	Measured result
Memory leaks	87 sections × 5 rounds, DOM nodes counted each round	+26,876 on the first round (construction), +0 on the four that follow
Exhausted storage	<code>setItem</code> replaced with a function that always throws	10 pages out of 10 stay operational
Degraded ≠ absent network	Requests held for 20 s, interception at context level	Answer from cache in 3,507 ms (threshold 3,500)
Spectrum reproducibility	Drawn twice, compared byte for byte	Identical
WebGL contexts	Viewer constructions over 10 consecutive molecules	From 7 to 1

Case	Method	Measured result
Corrupted history	Two Enter presses 500 ms apart	user,assistant instead of user,user,assistant,assistant
Text contrast	WCAG formula on every element with text of its own, 13 pages and 87 sections	19,751 elements examined (it was 41): 1,069 defects surfaced, 1,069 corrected , 0 recorded as a value that must not grow
Immediate cancellation	Stop pressed at 1.5 s, state sampled every second	Send button available from the 1st second (it was the 10th)

5. How to add a bench

Minimal structure

```

/* A comment explaining WHAT this bench demonstrates and why it is needed. */
const { chromium } = require('playwright-core');
const CHROME = '/opt/pw-browsers/chromium-1194/chrome-linux/chrome';

(async () => {
  const b = await chromium.launch({ executablePath: CHROME, args: ['--no-sandbox'] });
  // serviceWorkers:'block' if the bench modifies files: without it the SW
  // might serve a cached copy and the old file would be measured
  const ctx = await b.newContext({ serviceWorkers: 'block' });
  const pg = await ctx.newPage();
  const err = [];
  pg.on('pageerror', e => err.push(e.message));

  let ok = 0, ko = 0;
  const att = (d, atteso, avuto) => {
    if (String(atteso) === String(avuto)) { ok++; console.log(' ✓ ' + d + ' → ' + avuto); }
    else { ko++; console.log(' x ' + d + '\n      expected: ' + atteso + '\n      got:      '
+ avuto); }
  };

  await pg.goto('http://127.0.0.1:8899/index.html', { waitUntil: 'load', timeout: 60000 });
  await pg.waitForTimeout(2500);

  // ... measurements ...

  att('no JS error', 0, err.length);
  await b.close();
  console.log('\n' + (ko ? 'x ' + ko + ' FAILED, ' : '') + ok + ' passed');
  process.exit(ko ? 1 : 0); // the exit code is what counts
})();

```

The name is then added to the appropriate family in `tools/genera-evidenza.js`.

The rule that matters

A bench that measures nothing passes. It is the most insidious way to obtain a false guarantee: no alarm, no failure, and no verification.

Therefore **every bench that simulates a condition must count how many times the simulation actually fired, and fail if that count is zero.**

Traps actually met while building the battery:

Trap	How it showed itself
<code>page.route</code> does not intercept Service Worker requests (<code>context.route</code> is needed)	The "degraded network" test measured a healthy network and concluded in 16 ms instead of 3,500
Probing exhausted <code>localStorage</code> with a one-byte write	After saturation one byte always finds room: the bench "passed"
Looking for <code>[data-tab]</code> where navigation uses <code>[data-s]</code>	Zero tabs found, therefore zero measurements — and zero failures
The ticks of an axis on <code><canvas></code> are not text	They must be collected by intercepting <code>fillText</code> ; and the bottom-most line is the axis <i>title</i> , not a tick
A probe installed on a library loaded afterwards	It is replaced along with the library: zero calls counted while the code was working
A <code>var</code> inside an IIFE is not <code>window.something</code>	Assigning it from outside creates a different variable

6. Coverage gaps

Declared. The coverage reported in `08-Traceability-Matrix.md` measures **how many requirements have a bench**, not how many lines of code are executed: they are two different things and must not be confused.

Gap	Current situation	Recommendation
Code coverage	Measured: 49.79 % of statements over the widest path. What stays outside is whole-battery coverage and branch coverage: an <code>if</code> entered from one side only counts as covered	Extend collection to every bench, and move from statement coverage to branch coverage
Accessibility	Automated over 13 pages and all 87 sections: WCAG contrast, accessible names, labels, alternative text, heading hierarchy. Text inside SVGs and over gradients stay outside, and are counted on every run	Add <code>axe-core</code> alongside, for the rules this bench does not implement (ARIA roles, tab order, focus management)
Security	<code>verifica-sicurezza</code> runs 9 checks over 235 tracked files and covers SEC-01, SEC-03, SEC-05, SEC-06; SEC-04 is covered by <code>verifica_guida</code> . SEC-02 remains indirect : see <code>docs/09 D-03</code>	Observe the network traffic during real use, the only direct verification of SEC-02
Browsers other than Chromium	No automatic test on Firefox or WebKit	Extend the main benches to <code>webkit</code> , where the differences on IndexedDB and Service Worker are greatest
Performance	Manual cross-device testing	Automatic measurement of first-paint time

Gap	Current situation	Recommendation
Visual regression	Absent	Screenshot comparison for the charts, which are the heart of the product

7. Acceptance criteria

A version is not published if even one of these is unsatisfied.

- ❑ `node tools/genera-evidenza.js` closes with **0 failures**
- ❑ `tools/verifica-farmaci.js`: no defect, every deviation recorded with its reason
- ❑ `verifica_guida`: no promise in the documentation without a counterpart
- ❑ No non-network JavaScript error on any page
- ❑ Evidence report regenerated against the commit being published

Document updated to version `bsi-v172`.